INCH-POUND

MIL-M-49361C(CR)
26 May 1992
SUPERSEDING
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8 November 1989

MILITARY SPECIFICATION

MAGNIFIER, TAPERED FIBER OPTIC
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Magnifier, Tapered Fiber Optic, Viewer, herein known as "the magnifier" which is part of Viewer, Driver's Night Vision AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-O-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

STANDARDS

MILITARY

MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-C-771240	-	Magnifier, Tapered Fiber Optic, Viewer

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The magnifier is bonded to the Image Intensifier Tube and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The magnifier shall be constructed in accordance with SM-C-771240 and as specified herein.

3.3 Qualification. Unless otherwise specified by the contract or purchase order, Viewer Assemblies furnished under this specification shall be products approved as qualified products (see 3.11.3 and 6.3).

3.3.1 Initial production testing. The supplier shall furnish and test assemblies as specified in the contract (see 6.2.b).

3.4 Parts, materials and processes. Parts, materials, and processes shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Magnification. An axial image at the small end of the magnifier shall be 1.9, ± 0.04 times larger at the large end.

3.5.2 Distortion. The distortion shall be between 3 percent pincushion and 4 percent barrel.

3.5.3 Blemishes. Blemishes (see 3.11.7) shall not exceed the numbers specified in Table I. Zone 1 is the central 25 millimeters (mm) diameter; zone 2 is the central 40mm diameter less zone 1; and zone 3 is the remainder of the area out to a 48mm central diameter. In zone 1 the width of a continuous chicken wire (see 3.11.8) line shall not exceed 1 fiber but may be 3 fibers if interrupted with transparent areas equal to the opaque area in a regular cycle along the boundary between the multifibers. In zones 2 and 3, the width of the continuous chicken wire shall not exceed 2 fibers but may be 4 fibers if interrupted with equal length transparent areas.

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TABLE I. Blemish excluding chicken wire.

Size (inches	Number Zone 1	Number Zone 2	Number Zone 3
0.010 or larger	None	None	None
0.005 to 0.010 inclusive	5	10	10
0.003 to 0.005 inclusive	15	25	25
	Max % of Zone area	Max % of Zone area	Max % of Zone area
Smaller than 0.003	1%	1%	1%

3.5.4 Centering. On the small diameter face, the point which is on the mechanical axis of the taper (as defined by the large diameter) shall be imaged within a 0.040-inch diameter circle centered on the large face.

3.5.5 Deviation. A straight line image on the small diameter shall be imaged as a straight line within 1 mm on the large diameter.

3.5.6 Numerical aperture. The numerical aperture of the fiber optics shall be 1.0.

3.5.7 Fiber spacing. The spacing of fibers on the large end shall be 13 micrometers maximum.

3.5.8 Brightness distribution. The larger end of the magnifier is finished so as to produce the distribution of brightness as specified in Table II, with a diffuse input. Field angle is measured perpendicular to axis of the magnifier.

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TABLE II. Brightness distribution.

Zone	Exit Angle	Brightness
Central Zone out to 5 mm	+41 to -41° all directions	36% average 25% minimum
Intermediate Zone 5 to 15 mm radius from axis	+41° outward to -33° inward and ±40° sagittal	32% average 25% minimum
Outer Zone from 15 mm radius out to 23.9 mm	+40° outward to -30° inward and +35° sagittal	30% average 20% minimum
Flat Zone	+40° outward to -30° inward and -35° sagittal	25% average 15% minimum

3.6 Environmental characteristics.

3.6.1 High temperature. The magnifier shall withstand storage temperature of +68°C without damage or degradation of performance (see 3 11.1 and 3.11.5.)

3.6.2 Low temperature. The magnifier shall withstand storage temperature of -51°C without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.3 Temperature shock. The magnifier shall withstand cycles of sudden changes between -51°C and +68°C without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.4 Vibration. The magnifier shall withstand vibration equivalent to that encountered in tracked vehicles, during 5000 miles of operation without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.5 Shock.

3.6.5.1 Basic shock. The magnifier shall withstand a basic shock 40 g's (see 3.11.6) at 18 milliseconds duration without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.5.2 Weapon shock. The magnifier shall withstand a simulated weapon firing shock of 100 g's at 1.5 milliseconds duration, one half sine pulse, without damage or degradation of performance (see 3.11.1 and 3.11.5).

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3.7 Treatment and painting. Unless otherwise specified, the metallic parts of the magnifier shall be given a treatment and painting per MIL-STD-171.

3.8 Marking. The magnifier shall be marked and identified in accordance with MIL-M-13231.

3.9 Cleanliness. The magnifier shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

3.10 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any references specifications or drawings when inspected in accordance with Section 4. Parts and subassemblies shall exhibit none of the defects categorized in MIL-O-13830.

3.11 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.11.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.11.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.11.3 Qualified products. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.11.4 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

3.11.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.11.6 "g". "g" is a unitless measure of the quantity of units of force associated with the earth gravity that a object is subjected to.

3.11.7 Blemishes. Non-transmitting of fiber when viewed from the large end with a lambertian source at the small end.

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3.11.8 Chicken wire. Non-transmitting interrupted or continuous boundary area between multifibers when viewed from the large end with a lambertian source at the small end.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Initial production tests (4.3)
b. Quality conformance inspection (4.4)
c. Packaging inspection (4.7)

4.3 Initial production testing. Unless otherwise specified (see 6.2.b) ten magnifiers manufactured with production tooling shall be selected from the first production lot and provided for IPT. Following successful completion of the inspections specified in 4.3.1, each IPT magnifiers shall be subjected to all the tests in Table IV, which may be performed in any order. Failure of any tests shall be cause for rejection of the IPT.

4.3.1 Inprocess inspection. Prior to assembly, all materials, parts, processes and assemblies shall be examined for defects in accordance with the requirements of Table III.

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TABLE III. Inprocess inspection

Inspection	Requirement Paragraph	Inspection paragraph
Parts & materials	3.4	MIL-P-11268
Treatment & painting	3.7	MIL-STD-171
Marking	3.8	MIL-M-13231
Cleanliness	3.9	MIL-O-13830
Workmanship	3.10	MIL-O-13830

TABLE IV. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Magnification	3.5.1	4.5.1
Distortion	3.5.2	4.5.2
Blemishes	3.5.3	4.5.3
Centering	3.5.4	4.5.4
Deviation	3.5.5	4.5.5
Numerical aperture	3.5.6	4.5.6
Fiber spacing	3.5.7	4.5.7
Brightness distribution	3.5.8	4.5.8
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.3.2 Disposition of IPT Samples. Disposition of IPT samples shall be in accordance with the contract or purchase order (see 6.2.b).

4.4 Quality conformance inspection.

4.4.1 Examination. Each magnifier shall be examined as specified in 4.3.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Group A. Magnifiers that have passed the examination specified in 4.4.1 shall be tested in accordance with Table V in the order listed. Failure of any test shall be cause for rejection of that unit. Magnifiers which pass Group A inspection shall be formed into lots by the contractor.

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TABLE V. Group A inspection.

Inspection	Requirement Paragraph	Test Paragraph
Magnification	3.5.1	4.5.1
Distoration	3.5.2	4.5.2
Blemishes	3.5.3	4.5.3
Centering	3.5.4	4.5.4
Deviation	3.5.5	4.5.5
Numerical aperture	3.5.6	4.5.6
Fiber spacing	3.5.7	4.5.7
Brightness distribution	3.5.8	4.5.8

4.4.2.2 Group B. None required for this specification.

4.4.2.3 Group C. This inspection shall consist of the tests specified in Table VI and shall be performed on lots that have been subjected to and have met Group A inspection. Sampling shall be in accordance with contract or purchase order (see 6.2.c).

TABLE VI. Group C test schedule.

Test	Requirement Paragraph	Test paragraph
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2.d).

4.4.2.3.3 Disposition of Group C samples. Items which have passed Group C tests and are not damaged (see 3.11.1) may be shipped against the contract.

4.5 Test methods. Unless otherwise specified all tests shall be performed at ambient temperature.

4.5.1 Magnification. This test is performed concurrently with the distortion test given in 4.5.2. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

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4.5.2 Distortion. Set up test equipment as shown in Figure 1. Back illuminate the test target. Place the magnifier such that its optical axis is coincident with the optical axis of the microscope. Traverse the microscope and verify that the 12.5 mm extremes of the test target are visible. Failure to see both 12.5 mm extremes constitutes failure to meet the requirements of 3.5.2. Next, starting at the +12.5 mm field height, record the travel distance of the microscope versus field height in 5 mm increments until -12.5 mm field height is obtained. Consider the ± 2.5 mm field height readings and obtain the average. From this magnification, M shall be calculated using:

$$M = \frac{\text{Height of image (Average)}}{2.5 \text{ mm}}$$

Percent distortion, DF, at any field height is calculated as:

$$DF = (F/MF - 1) \times 100$$

Where:

F = test pattern height in mm

$\bar{F}$ = average of the measured image heights in mm

M = on-axis magnification

Failure to meet the requirements of 3.5.1 and 3.5.2 shall constitute failure of this test.

4.5.3 Blemish. The output of the magnifier is examined for blemishes (see 3.11.7) and chicken wire (see 3.11.8) with a 10 power eye loupe. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Centering. The magnifier's large diameter is rotated on its mechanical axis in a fixture that presents on this axis a point of light to the small end of the taper. The position of the image of this point of light on the large end is measured. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Deviation. A straight edge is placed on the small end and the departure from a straight edge is measured with a comparator. Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Numerical aperture. Samples of the material used in each melt shall be examined for numerical aperture. Failure to meet requirement of 3.5.6 shall constitute failure of this test.

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4.5.7 Fiber spacing. Samples of the material used in each melt shall be examined for fiber spacing. The spacing shall be measured with a traveling microscope. Failure to meet requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Brightness distribution. The brightness distribution shall be measured by rotating an imaging photometer so as to measure the angles and field points as specified in 3.5.8 (see Figure 2). Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.6 Environmental. The magnifier shall be retested to meet the requirements of 3.5.1, 3.5.2., 3.5.3, and 3.5.8 after the completion of each environmental test (see 3.11.4). Failure to meet these requirements shall constitute failure as defined in degradation of performance. Temperature chambers shall maintain specified temperatures within ± 2.0°C.

4.6.1 High temperature. The magnifier shall be subjected to the test of Method 501.1, Procedure I of MIL-STD-810, except that the temperature of Step 2 (highest storage temperature) shall be +68°C and Steps 4 and 5 shall be deleted. Step 6 stabilization period shall be a minimum of two hours. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, 3.5.8, and 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The magnifier shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -51°C. Steps 2 and 6 stabilization period shall be a minimum of two hours, and Steps 4 and 5 shall be deleted. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, 3.5.8, and 3.6.2 shall constitute failure of this test.

4.6.3 Temperature shock. The magnifier shall be subjected to the test of Method 503.1, Procedure I of MIL-STD-810, except the high temperature chamber shall be maintained at +68°C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Immediately after each cold cycle, the magnifier shall be examined for damage (see 3.11.1). Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, 3.5.8, and 3.6.3 shall constitute failure of this test.

4.6.4 Vibration. The magnifier shall be subjected to MIL-STD-801, Method 514.2, Procedure VIII, Table 514.2-6 tracked vehicles, limited to one sweep in the vertical axis only. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, 3.5.8, and 3.6.4 shall constitute failure of this test.

4.6.5 Shock. The magnifier shall be subjected to the test of MIL-STD-810, Method 516.2. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, 3.5.8, and 3.6.5 shall constitute failure of this test.

a. 40g's at 18 milliseconds duration, Figure 516.2-1, shock Procedure I.
b. Weapon firings, 100g's using procedure IV except at 1.5 milliseconds (ms) duration, Figure 516.2-2.

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4.7 Inspection for packaging. Packaging shall be inspected to determine compliance with section 5.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The magnifier covered by this specification is intended for use in the Viewer, Driver's Night Vision AN/VVS-2(V), an optical viewing device capable of directing the line of sight to the driver's eyes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Quantity and schedule of initial production testing and disposition of IPT samples.

c. Sampling plans for Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.

d. Necessary actions by the contractor in the event of a Group C failure (see 4.4.2.3.2).

e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

f. MIL-STD-810C is used for environmental tests of section 4.

g. Packaging requirements.

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6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.11.

6.5 Subject term keyword listing.
Magnifier
Viewer Driver's

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CR

Review Activity:
Army - ME, TE

User Activity
Army - AT, MU

Preparing Activity:
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Project: 5855-A365

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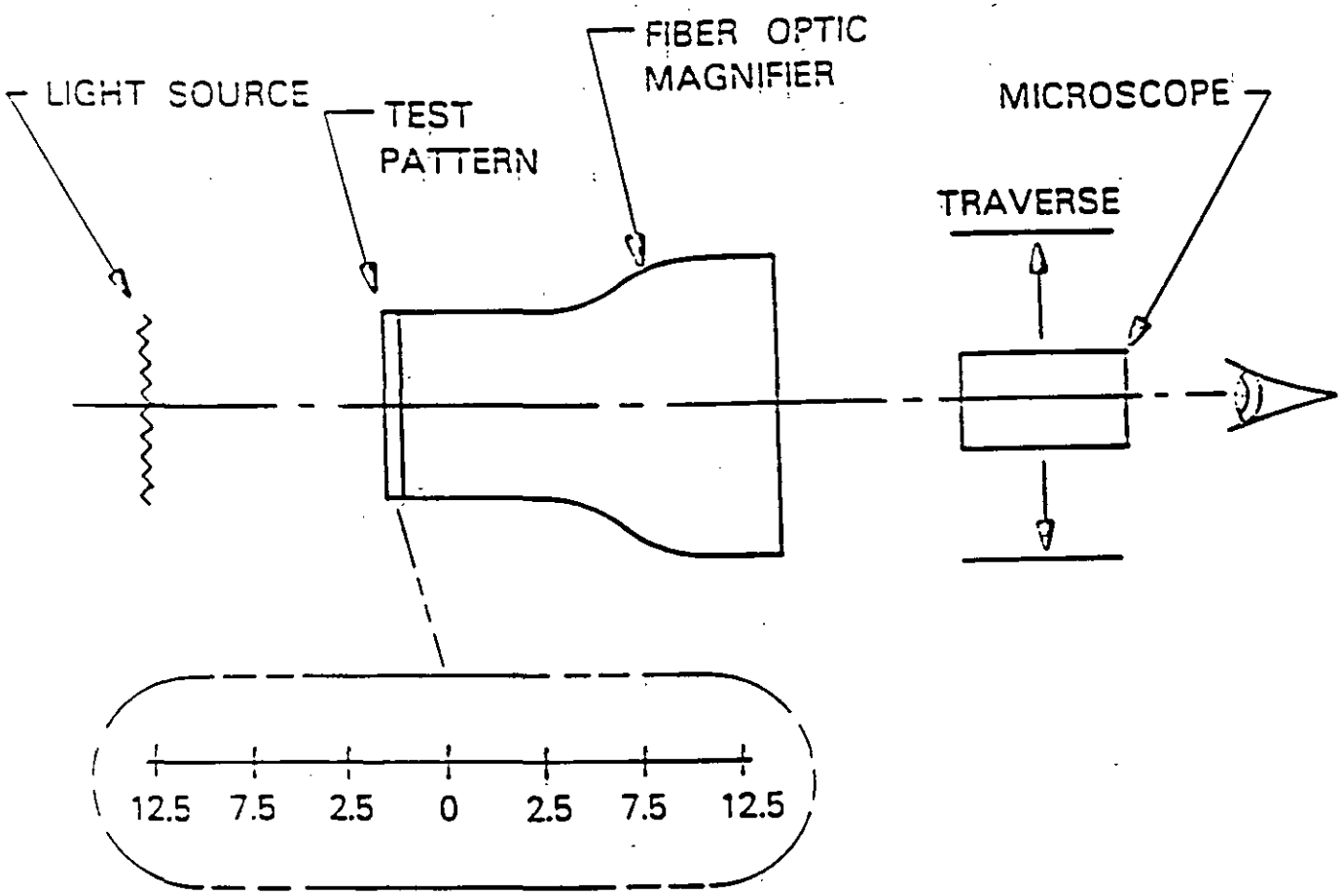


FIGURE 1. FIBER OPTIC MAGNIFIER TESTS DIAGRAM

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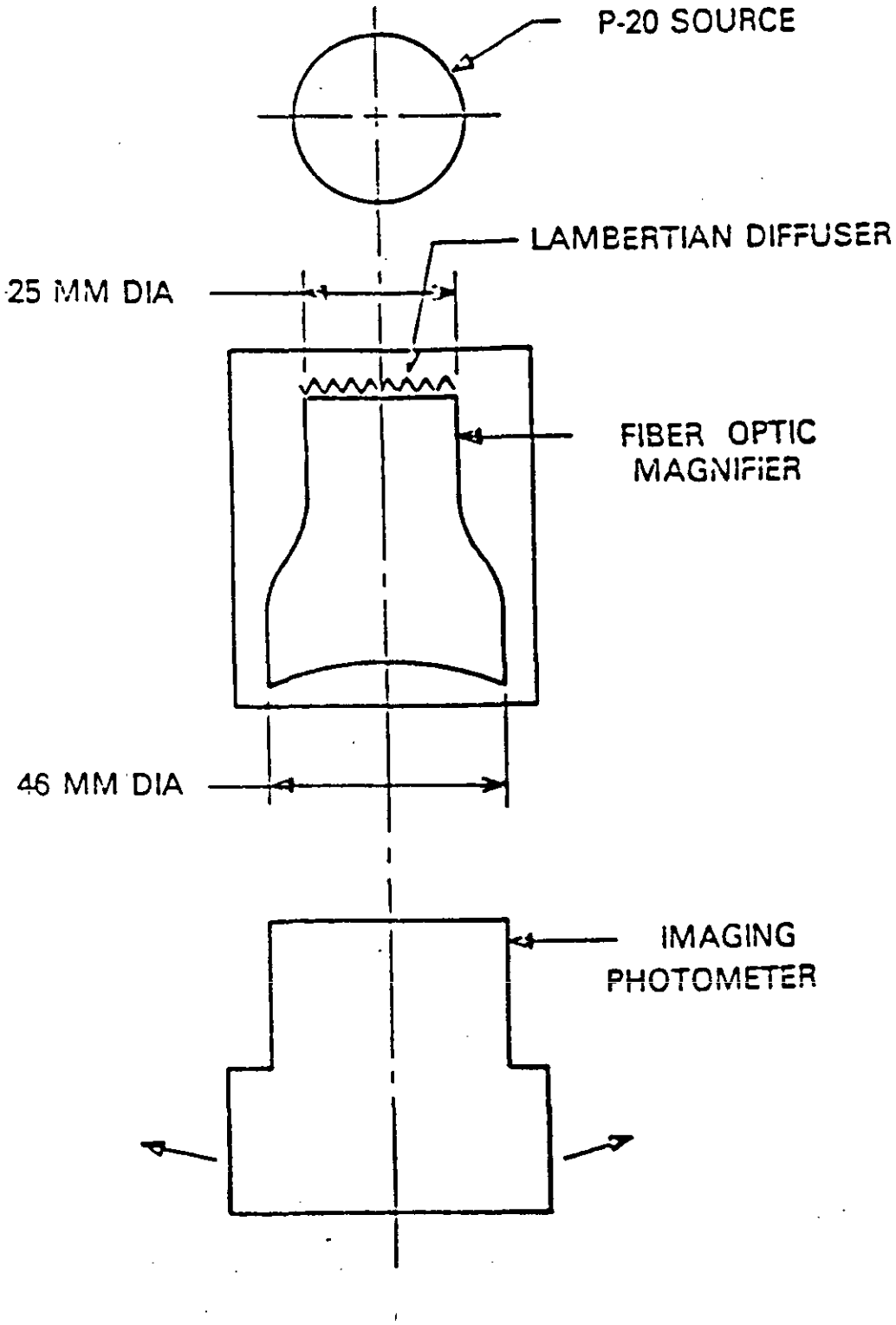


FIGURE 2. BRIGHTNESS DISTRIBUTION TEST DIAGRAM

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-49361C(CR)	26 May 1992

3. DOCUMENT TITLE
MAGNIFIER, TAPERED FIBER OPTIC VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908)532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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